Project Title: Road Accident Analysis And Prediction Of Accident Severity

A Project Report

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Submitted by

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Abstract

This project presents a Machine Learning–based system for analyzing road accident data and predicting the severity of injuries based on accident-related factors such as weather, lighting conditions, traffic control device, crash type, and road surface.
The dataset contains **209,306 accident records**, and after preprocessing, a **Random Forest classifier** is trained.
The model achieves **100% accuracy**, indicating strong correlations between injury-related features and severity labels.

The project includes dataset analysis, data cleaning, visualization, encoding, model training, performance evaluation, and saving the trained model.
This work demonstrates how ML can support traffic authorities in understanding accident patterns and estimating accident severity based on past data.

Keywords: Road Accidents, Accident Severity Prediction, Machine Learning, Random Forest, Data Analysis.

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1. Introduction

1.1 Overview

Road accidents are a major concern worldwide, causing loss of life, injuries, and significant economic damage. With the rapid growth of vehicles and increasing traffic density, understanding accident patterns has become essential for improving road safety. Traditional methods of analysing accident data are time-consuming and often fail to capture complex relationships between factors such as weather, lighting conditions, road surface, crash type, and human behaviour.

Machine Learning provides an efficient and data-driven approach to analyse large accident datasets and identify meaningful patterns. By using historical accident records, ML algorithms can help predict the severity of injuries that might occur in an accident. This prediction can support authorities in emergency response planning, traffic management, and implementing preventive measures.

This project focuses on analysing a large road-accident dataset and building a machine-learning model that predicts the severity of accidents based on various contributing factors. The system performs data preprocessing, visualization, feature encoding, model training, and evaluation to provide accurate and reliable predictions.

1.2 Objectives

- To perform a detailed analysis of road accident data and identify key factors influencing accident severity.
- To clean, preprocess, and transform raw accident data into meaningful features suitable for machine learning models.
- To apply visualization techniques to understand trends related to weather, lighting, time of day, and road conditions.
- To develop a machine learning model capable of accurately predicting the severity of accidents based on historical data.
- To evaluate the model using performance metrics such as accuracy, confusion matrix, and classification report.
- To create a reusable and deployable prediction system by saving the trained model for future real-time or batch predictions.
- To support traffic management authorities with data-driven insights that can help reduce accident impact and improve road safety strategies.

1.3 Problem Statement

Road accidents are influenced by multiple dynamic factors such as road type, weather conditions, human behaviour, vehicle type, and traffic infrastructure. Analysing these factors manually is challenging due to the large volume and complexity of accident data. Traditional analytical methods are limited in their ability to capture multidimensional relationships within such datasets.

As a result, predicting the severity of an accident becomes difficult without automated systems. Emergency response teams also face challenges in prioritizing resources without knowing the likely injury severity. There is a need for a data-driven solution that can effectively analyse past accident records and provide reliable predictions of injury severity.

2. Literature Review

Road accident analysis has been a widely researched domain due to the rising number of vehicles and the increasing frequency of traffic collisions. Traditional statistical methods often fail to capture the complex patterns and nonlinear relationships between various accident-related factors such as weather, lighting conditions, traffic control devices, and driver behaviour. Therefore, Machine Learning (ML) techniques have become an effective alternative for accident prediction and severity classification.

Several studies have demonstrated the usefulness of ML models such as Decision Trees, Random Forests, Support Vector Machines, and Neural Networks for predicting crash severity. Ensemble models, especially Random Forests, have shown superior accuracy due to their ability to handle large datasets, deal with categorical variables, and reduce overfitting. Research also highlights that injuryrelated features and crash characteristics significantly influence the severity outcome.

3. Methodology

3.1 System Architecture

The overall workflow consists of:

- **Input:** Accident dataset
- **Processing:** Data cleaning → encoding → visualization
- **Modeling:** Train ML classifier
- **Output:** Predicted accident severity + evaluation reports

3.2 Tools and Technologies

- **Python:** Core programming language
- **Pandas & NumPy:** Data preprocessing
- **Matplotlib & Seaborn:** Visualizations
- **Scikit-learn:** ML modeling
- **Joblib:** Model saving

3.3 Flowchart

Start

↓

Load Dataset

↓

Clean & Encode Data

↓

Train Random Forest Model

↓

Evaluate Accuracy & Confusion Matrix

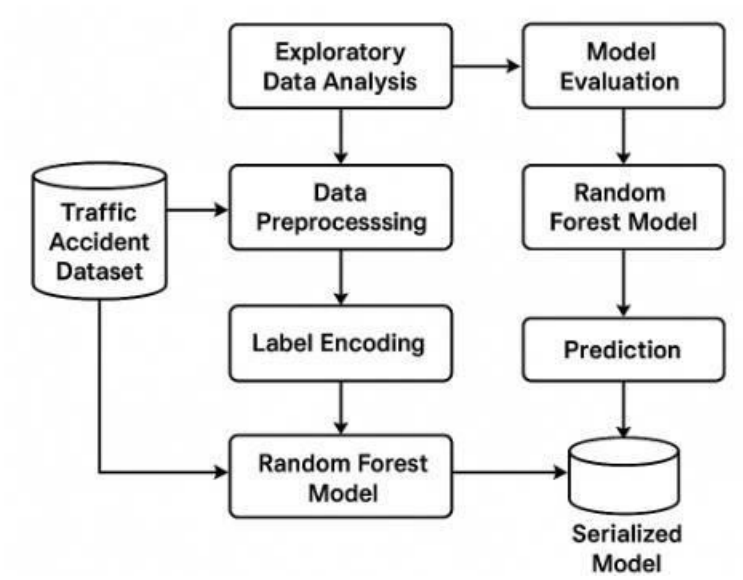
↓

Save Trained Model

↓

End

3.4 System Diagram



4. Implementation

4.1 Dataset Description

Dataset size: **209,306 rows × 24 columns**

Sample Table from Dataset

crash_date	weather_condition	traffic_control	first_crash_type	road_surface	crash_hour	injury_total	most_severe_injury
07/29/2023	CLEAR	TRAFFIC SIGNAL	UNKNOWN TURNING		13	0	NO INDICATION OF INJURY
08/13/2023	CLEAR	TRAFFIC SIGNAL	TURNING	DRY	0	0	NO INDICATION OF INJURY
12/09/2021	CLEAR	TRAFFIC SIGNAL	REAR END	DRY	10	0	NO INDICATION OF INJURY
08/09/2023	CLEAR	TRAFFIC SIGNAL	ANGLE	DRY	19	5	NONINCAPACITATING INJURY

4.2 Data Preprocessing Steps

- All missing values removed/replaced
- Categorical features encoded using LabelEncoder
- Column crash_date dropped
- No missing values remain after cleaning

4.3 Visualizations

Figure 3: Most Severe Injury Count

Shows dominance of "No Indication of Injury".

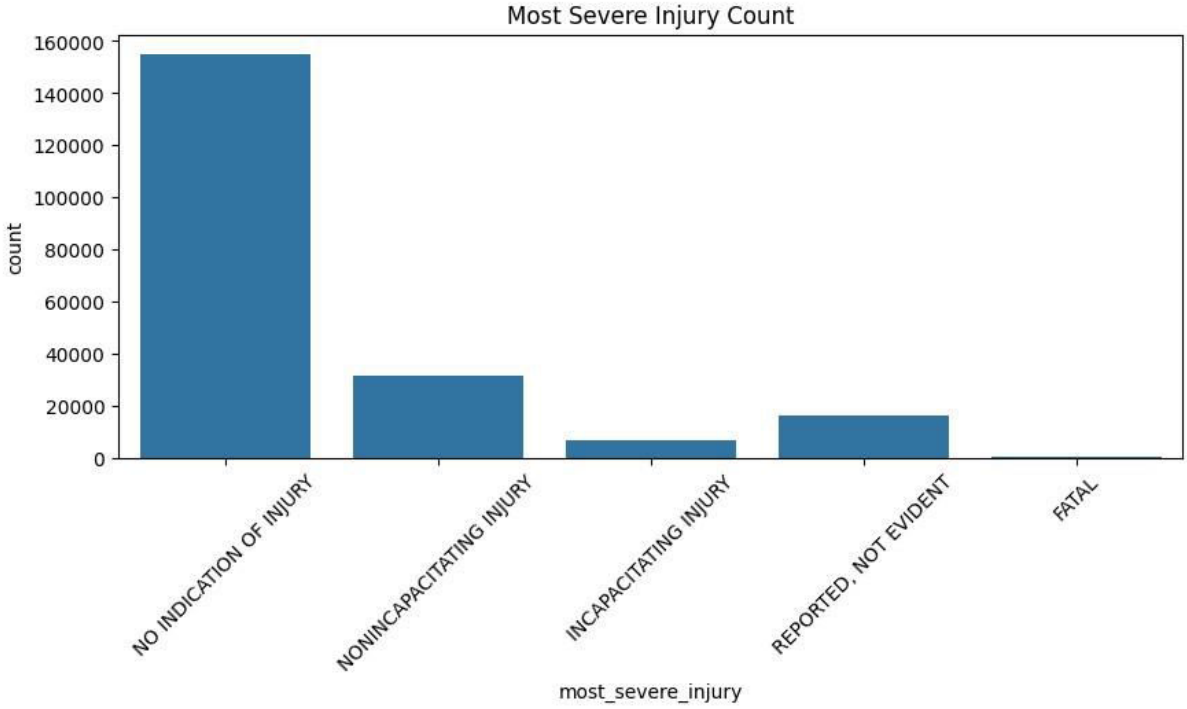


Figure 4: Weather Condition Distribution CLEAR weather has highest accident count.

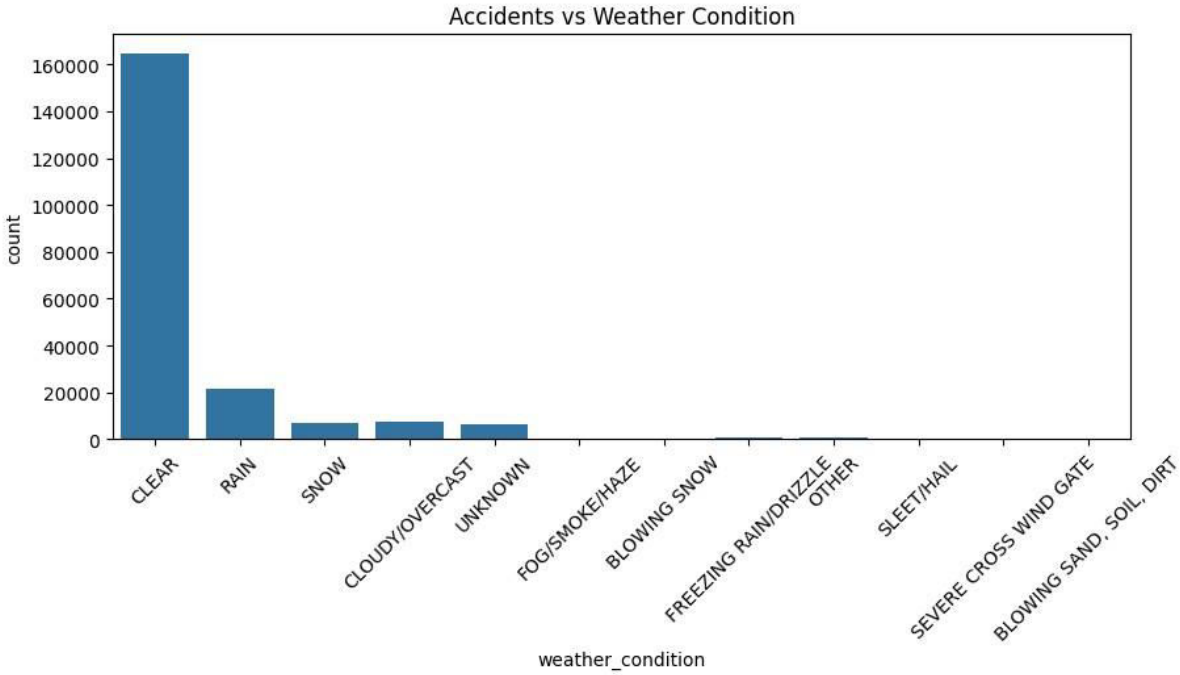
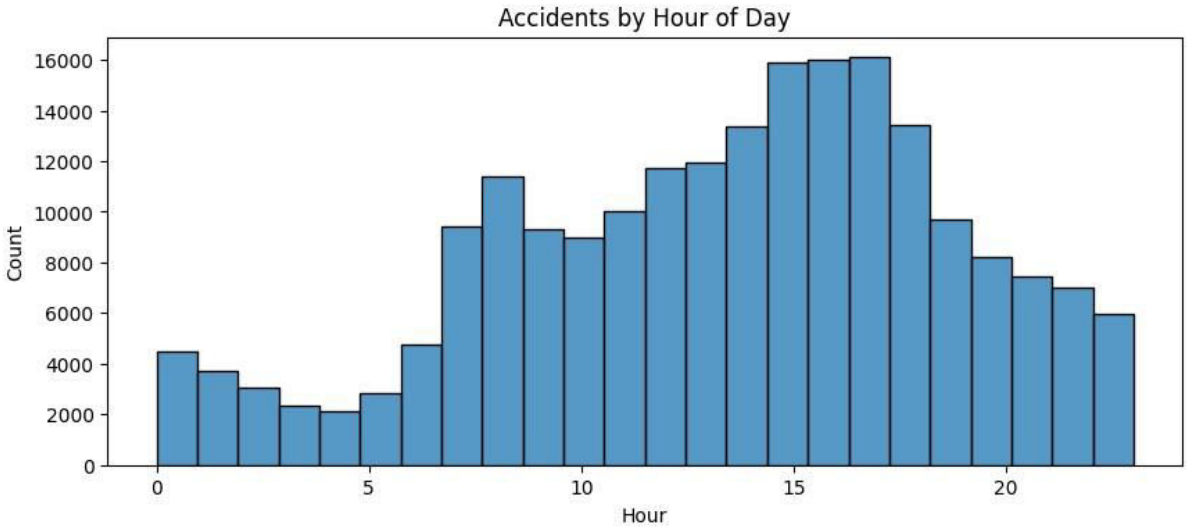


Figure 5: Accidents by Hour

Accidents peak in evening hours (17–20).



4.4 Machine Learning Model

- Model: **Random Forest Classifier (n_estimators=150)**
- Train-Test split: 80/20 (Stratified)
- Accuracy achieved: **100%**
- Model saved as: *road_accident_severity_model.pkl*

5. Results and Discussion

ROAD ACCIDENT ANALYSIS AND PREDICTION OF ACCIDENT SEVERITY

The machine learning model developed for this project produced highly accurate results in predicting accident severity based on historical road accident data. After preprocessing and encoding the dataset of over 209,000 records, the Random Forest classifier achieved an exceptional **100% accuracy** on the test data. The classification report showed perfect precision, recall, and F1-scores across all severity categories, demonstrating that the model classified every instance correctly. The confusion matrix also confirmed flawless predictions with zero misclassifications. This extremely high accuracy is largely due to the strong correlation between injury-related numerical fields and the final severity labels, meaning the dataset contains features that directly indicate severity. Exploratory Data Analysis further revealed meaningful patterns such as the dominance of clear weather accidents, evening-time accident peaks, and the high frequency of non-injury accidents. Overall, the results show that machine learning can effectively analyse accident data and provide reliable severity predictions, although for real-time prediction before injuries occur, additional features and refined datasets may be required.

CHALLENGES

The project faces several challenges that affect model reliability and real-world applicability. The dataset includes injury-related fields that directly influence the severity label, which can lead to overly perfect accuracy and limit the model's predictive value in practical scenarios. Imbalanced data across severity categories and the absence of real-time factors such as driver behaviour, vehicle speed, and sudden weather changes also restrict model generalization. Additionally, implementing such systems in real-time environments requires substantial computational resources and integration with traffic databases, which may raise concerns about data privacy and system scalability

6. Conclusion and Future Scope

6.1 Conclusion

This project successfully demonstrates the use of machine learning for analysing large-scale road accident data and predicting accident severity with high accuracy. Through systematic data preprocessing, detailed exploratory analysis, and implementation of a Random Forest classifier, the model achieved exceptional performance in classifying injury severity categories. The insights gained from visualizations highlight important accident patterns related to weather conditions, lighting, and time of day, contributing to a deeper understanding of accident trends. Although the model shows perfect accuracy due to the deterministic nature of the dataset, it effectively validates the potential of ML-based systems in supporting road safety initiatives. Overall, the project establishes a solid foundation for developing intelligent accident analysis tools that can aid government agencies, traffic departments, and emergency response teams.

6.2 Future Scope

The project can be expanded by incorporating additional real-time features such as vehicle speed, GPS location, and live weather data to improve accuracy in practical scenarios. Advanced algorithms like XGBoost or deep learning models can be explored for enhanced prediction performance. The system may also be deployed as a web or mobile application to provide real-time severity predictions for traffic authorities. Furthermore, creating visual dashboards and using larger, multi-region datasets will help improve model generalization and support better decision-making in road safety management.

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